Part 5 presenter script: slides 34–41 (5-minute cut)

Visualisation Cases and Defender Extension · Srinjoy Chakraborty · CS637, Stealthy Computational Delay Attacks on Control Systems (Nauta, Sandberg & Maggio, ICCPS 2025)

SLIDE	CONTENT	TIME
34	Divider	0:00–0:05
35	Furuta: three Table 1 HOLD attacks, plus visualisation results	0:05–1:05
36	Furuta: horizon and detector validation	1:05–1:40
37	Tank: detector comparison	1:40–2:15
38	Tank: HOLD vs ZERO	2:15–2:50
39	Defender min–max formulation	2:50–3:35
40	Defender validation: missing horizon	3:35–4:15
41	Conclusion	4:15–4:45

> There is no live demo; stay on the slides throughout. The script is about 580 spoken words. The last version had about 1,170 words and took you about 10 minutes, so this one should run just under 5 minutes. Italic grey lines are stage cues. The small grey brackets after symbols are silent memory aids, so do not read them aloud. Notation matches the edited deck: S_{χ^2} is the geometric statistic and $\chi^2_{.99}(4) = 13.277$ is its limit.

Slide 34 · Visualisation Cases and Defender Extension

0:00–0:05

Thank you, Abhishek. I'll replay the paper's attacks on both plants, and then look at the problem from the defender's side.

Slide 35 · Furuta: the three published HOLD attacks

0:05–1:05

Point to the parameter line, trace the curves, then the screenshot and table.

These are the three Table 1 schedules: ℓ*(attack length)* = 25, $\hat{p}_m$*(nominal miss probability)* = 0.25, $\bar{a}$*(watchdog limit)* = 4. They were optimised for **HOLD**, and we replay each one under both policies. In Equation 4, only Δ*(actuator-memory block)* changes.

As the figure shows, the HOLD curves fall away in the final periods, while **ZERO** stays flat. We also built an interactive visualisation, shown in the screenshot, that replays these schedules on the authors' exact matrices. It shows the pendulum falling to **22°** under HOLD but never passing its initial 0.57° under ZERO, and it shows the attack as bursts of three misses that sit exactly at the binomial limit.

In the table, ω*(extra terminal cost vs. no attack)* under HOLD is **0.144, 0.078 and 0.088**. Under ZERO it is about minus 7×10^{-6}, slightly better than nominal, because a held stale command keeps pushing the wrong way. On the right, all three schedules pass the watchdog and binomial tests but fail the geometric test: S_{χ^2}*(geometric χ^2 statistic)* = **187, 86 and 95**, against $\chi^2_{.99}(4)$*(critical value)* = 13.277.

Slide 36 · Furuta: horizon and detector validation

1:05–1:40

Point to the ω_{ZERO} column, then the "Extra" row.

These are the paper's ZERO-optimised schedules under the combined detector. As the table shows, ω_{ZERO}*(ω under ZERO)* rises from 2.7×10^{-5} at $\ell = 5$ to 8.5×10^{-5} at 15, then falls back at 25. **A longer attack does not**

mean more terminal damage. All three pass, with S_{χ^2} up to 12.8. The extra 23-period schedule passes the binomial test only, with $S_{\chi^2} = 126$, and it is hundreds of times more damaging under HOLD.

Slide 37 · Quadruple tank: detector comparison

1:40–2:15

Point to the curves, then the first two table rows.

For the stable quadruple tank, we enumerate all **1,024** ten-period schedules, so these are exact optima. As the figure shows, y_2*(lower-tank output)* deviates most under the binomial-only optimum: $\omega = 0.0032$, but $S_{\chi^2} = 339$. Adding the geometric constraint gives 0.0021 with $S_{\chi^2} = 10.7$. That is a **34.4% reduction**, which supports the paper's Figure 3.

Slide 38 · Quadruple tank: policy comparison

2:15–2:50

Point to the paired bars, then the equation.

Our extension applies ZERO to the tank, meaning zero input *deviation*, not pumps off. As the bar chart shows, damage **increases**, by **2.1 to 2.9 times**. The pendulum is unstable, so a stale command hurts it. The tank is slow, so the held command is still roughly right. The equation, $\omega_y = ¼\ \omega_{REP}$*(output cost = ¼ of REP's state cost)*, fixes the units.

Slide 39 · Defender study: formulation

2:50–3:35

Trace the equation from the inside out.

Now the defender. As the equation shows, $d^* = arg\ min_d\ max_{\ell \in \mathscr{L}} \omega^*$*(d = defender choice; $\mathscr{L}$ = tested horizons; ω^* = attacker's best ω)*, subject to $FA \le \varphi$*(session false-alarm rate ≤ limit)*. The attacker picks the worst stealthy schedule, we take the worst horizon, and the defender picks the policy, detector, window and α*(significance level)* that make it smallest. Without the false-alarm limit, the defender would just choose the tightest detector. The table shows the status: the replays are done, the binomial sweep is partial, and the other detectors are planned.

Slide 40 · Defender validation: horizon coverage

3:35–4:15

Trace the top curve, then point to 6.56.

As the figure shows, the maximum ω against the window n_w*(binomial window length)* is **not monotone**. The key number is in the table. The saved maximum is 0.022, but the $\ell = 25$ attack we replayed reaches 0.144, which is **6.56 times larger**. Those runs failed because of the Gurobi licence limit. **So a partial maximum is never a safety guarantee.**

Slide 41 · Conclusion

4:15–4:45

To conclude, deadline misses alone are an attack surface. We reproduced the paper's curve, showed that its Table 1 attacks fail the geometric test, found that ZERO helps the pendulum but hurts the tank, and formulated a defender study. The main limits are the terminal-only cost and the missing $\ell = 25$ runs. The takeaway is that timing, policy and detection must be designed together. Thank you.

Backup: quick answers for Q&A

not timed

Why does the geometric test reject when binomial passes?	The binomial test counts misses per window. The geometric test checks how they are grouped: bursts of exactly 3 are very rare at $\hat{p}_m = 0.25$ (about 0.13 expected, 5 observed).
Where does 13.277 come from?	$\chi^2_{.99}$ with $\bar{a} = 4$ degrees of freedom at $\alpha = 0.01$. Abhishek's 11.345 is the same quantity with $\bar{a} = 3$.
What does the visualisation show?	It is an offline dashboard (`visualisations/furuta_table1_dashboard.html`). It shows HOLD vs ZERO side by side for the three Table 1 schedules, the miss strip, the binomial window counts, the geometric observed-vs-expected bars, and a detector audit.
Is ZERO safer?	Not universally. It helps the pendulum and hurts the tank, and neither result is a worst case because the attacker was not re-optimised for ZERO.
Is 34.4% a detector ranking?	No. It compares constraint sets; a fair ranking needs equal session false-alarm rates.
Why is the $\ell = 25$ data missing?	Gurobi error 10010: the model exceeded the size-limited licence. 432 of 540 runs were saved.